WETLAND DEGRADATION ASSESSMENT

Assessment of ecological change within the selected analysis area

OBSERVATION PERIOD	2026-09-01 → 2026-09-03	INTERVAL	2 days
WETLAND CONTEXT	User-selected analysis area	RUN ID	run_20260903_section21_example
ANALYSIS FOOTPRINT	100.00% active area	SPATIAL REFERENCE	EPSG:32633

OVERALL WETLAND ASSESSMENT

NO MEASURABLE WETLAND DEGRADATION DETECTED

No measurable wetland degradation was detected from the available evidence during the observation period. Vegetation condition remained stable across the selected area. Spectral indicators showed zero measurable displacement. Surface-water change was evaluated separately, but quantitative water extent was unavailable for this run. Therefore, the available evidence does not establish physical wetland loss.

WETLAND-LEVEL INDICATOR	OBSERVED RESULT
Wetland Degradation (Hydro-Ecological)	Not available
Wetland Recovery (Hydro-Ecological)	Not available
Stable Classified State	100.00%
Vegetation Degradation (Supporting Evidence)	0.00%
Vegetation Recovery (Supporting Evidence)	0.00%
Surface-Water Loss (Contraction)	Not available
Continuous Spectral Shift (Mean CVA)	0.0000
Effective Analysis Area	100.00%

1. THE AREA WE ANALYZED

This assessment examines changes within the area selected for analysis. Vegetation condition, spectral indicators and surface-water evidence are compared between the two observation dates. The selected area defines where the assessment is performed. It should not automatically be interpreted as a legal or statutory wetland boundary.

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SPATIAL ATTRIBUTE	OBSERVED / CONFIGURED VALUE
Wetland context status	WETLAND_BOUNDARY_UNAVAILABLE
Wetland source	User-selected analysis area
Source dataset / boundary	User-defined analysis boundary
Selected analysis area	100.00%
Effective analysis footprint	100.00%
Manual exclusion footprint	0.00% (No exclusions configured)
Physical wetland loss pathway	NOT_ASSESSED

2. WETLAND CONDITION

Wetland condition is evaluated by integrating multiple independent evidence streams. Vegetation condition, continuous spectral dynamics, and surface-water presence are each assessed across the active analysis footprint. The table below outlines how each evidence stream informs the integrated wetland assessment.

EVIDENCE STREAM	OBSERVED FINDING	ROLE IN ASSESSMENT
Vegetation condition	Stable canopy state (100.00% persistent)	Tracks canopy structure and foliar condition class shifts.
Spectral change	Mean CVA magnitude: 0.0000	Measures continuous, sub-classification physiological canopy displacement.
Surface-water change	Quantitative water extent unavailable for this run	Independently evaluates wetting and drying trends.
Integrated assessment	**NO MEASURABLE WETLAND DEGRADATION DETECTED**	Synthesizes multi-stream evidence into the wetland degradation finding.

Evaluating wetland condition requires distinguishing transient canopy fluctuations from systemic hydro-ecological loss. Optical vegetation stress without concurrent hydrological contraction indicates vegetation-level dynamics, whereas concurrent vegetation stress and water loss provide evidence of wetland degradation.

3. VEGETATION EVIDENCE

Vegetation condition serves as an essential line of evidence in the wetland assessment. Canopy condition is classified at baseline (T1) and present (T2) into four discrete states: Healthy Vegetation, Moderate Stress, Severe Stress, and Background / Non-Vegetated. All proportions are reported as percentages of the effective analysis footprint.

CONDITION CLASS	BASELINE (T1) %	PRESENT (T2) %	NET CHANGE (%)
Healthy vegetation	100.00%	100.00%	0.00%
Moderate stress	0.00%	0.00%	0.00%
Severe stress	0.00%	0.00%	0.00%
Background / non-vegetated	0.00%	0.00%	0.00%

VEGETATION TRANSITION DYNAMICS

Transitions between observation dates trace the exact trajectory of vegetation shifts, separating movement toward higher stress (degradation pathways) from recovery or canopy loss.

TRANSITION PATHWAY	DEFINED TRANSITIONS	FOOTPRINT PROPORTION
Persistent classified state	H→H, M→M, S→S, B→B	100.00%
Movement toward poorer vegetation condition	H→M, H→S, M→S	0.00%
Movement toward better vegetation condition	M→H, S→M, S→H	0.00%
Canopy loss / non-vegetated transition	H→B, M→B, S→B	0.00%
Canopy emergence / colonisation	B→H, B→M, B→S	0.00%

Vegetation evidence summary: 100.00% of the active analysis area remained in a stable classified condition state. 0.00% followed vegetation degradation pathways, and 0.00% followed recovery pathways.

4. SPECTRAL EVIDENCE

Continuous spectral indices provide independent evidence of foliar and physiological condition shifts. Change Vector Analysis (CVA) integrates ΔNDVI, ΔGNDVI, and ΔNDRE in a multi-dimensional feature space to capture subtle canopy adjustments that occur within or across categorical classification boundaries. These metrics represent continuous radiometric evidence rather than standalone wetland degradation measures.

SPECTRAL INDICATOR	MEAN	MEDIAN	P95
CVA Magnitude (||ΔV||)	0.0000	0.0000	0.0000
ΔNDVI (ΔNDVI)	N/A	N/A	N/A
ΔGNDVI (ΔGNDVI)	N/A	N/A	N/A
ΔNDRE (ΔNDRE)	N/A	N/A	N/A

Spectral evidence summary: Mean CVA magnitude is 0.0000 across the analysis area, indicating zero measurable average radiometric displacement between baseline and present observations.

5. SURFACE-WATER EVIDENCE

Surface-water presence and moisture dynamics provide an essential hydrological line of evidence. Water presence is evaluated from the Normalized Difference Water Index (NDWI) using a calibrated threshold of 0.00. Where quantitative water extent cannot be derived from available observations, metrics are reported as 'Not available' to prevent misleading zero-loss assumptions.

HYDROLOGICAL METRIC	FOOTPRINT PROPORTION	PHYSICAL EXTENT	DEFINITION
Baseline Water Extent (T1)	Not available	Not available	Surface water detected at baseline observation
Present Water Extent (T2)	Not available	Not available	Surface water detected at present observation
Surface Water Expansion (Gain)	Not available	Not available	Newly emerged surface water / wetting
Surface Water Contraction (Loss)	Not available	Not available	Previously inundated surface becoming dry
NDWI Water Threshold	0.00	N/A	Calibrated Green/NIR NDWI zero-plane threshold

Hydrological evidence summary: Quantitative surface-water extent was unavailable for this run; hydrological presence could not be independently quantified.

6. WETLAND DEGRADATION ASSESSMENT

This section presents the primary analytical synthesis of the assessment. The evaluation synthesizes all available independent evidence streams—vegetation condition, continuous spectral shifts, and surface-water dynamics—to determine whether wetland degradation occurred within the selected analysis area between observation dates.

Distinction between Vegetation Degradation and Wetland Degradation:
Vegetation stress alone does not establish wetland degradation. A hydro-ecological wetland degradation signal is established only when concurrent evidence of both vegetation deterioration and surface-water loss is detected. Where surface-water dynamics are unavailable or stable, vegetation shifts represent vegetation-only condition responses (such as physiological stress or seasonal variation) rather than systemic wetland loss.

PRIMARY ASSESSMENT: NO MEASURABLE WETLAND DEGRADATION DETECTED

Available evidence does not indicate measurable wetland degradation within the selected analysis area during the observation period. Classified vegetation condition remained completely stable, and continuous spectral indicators confirmed zero average displacement. Surface-water extent was not quantitatively available for this run, so physical wetland loss is not established.

PHYSICAL WETLAND LOSS: NOT ESTABLISHED

Physical wetland loss is not established. In accordance with strict scientific safeguards, optical vegetation change alone is not interpreted as permanent physical wetland loss.

7. EVIDENCE FUSION

Evidence fusion cross-references vegetation transition pathways with surface-water dynamics. A composite hydro-ecological degradation signal requires concurrent evidence of both vegetation stress and surface-water loss, preventing single-sensor optical artifacts from triggering false alarms.

EVIDENCE CLASS	FOOTPRINT PROPORTION	PHYSICAL AREA	DEFINITION & MEANING
Stable landscape	100.00%	N/A	Persistent classified condition state across both observation dates
Hydro-ecological degradation	Not available	Not available	Concurrent vegetation degradation and surface-water loss
Hydro-ecological recovery	Not available	Not available	Concurrent vegetation recovery and surface-water gain
Vegetation-only degradation	0.00%	N/A	Vegetation condition decline without concurrent hydrological loss

8. DATA QUALITY & SPATIAL ACCOUNTING

Spatial accounting verifies that every spatial component is traceable from source imagery through the inclusion boundary and manual exclusions to the effective analysis footprint.

QUALITY CONTROL CHECK	STATUS	FINDING
Spatial grid alignment	PASS	T1 and T2 rasters verified on a common analysis grid without spatial distortion.
Footprint reconciliation	PASS	Selected analysis area fully reconciled with manual exclusions (100.00% accounted for).
Independent ground reference	UNAVAILABLE	No external ground-truth validation plots were supplied for this run.
Physical depletion guard	ENFORCED	Optical vegetation decline is prevented from being reported as physical wetland loss.

SPATIAL FOOTPRINT RECONCILIATION

FOOTPRINT COMPONENT	PROPORTION OF SELECTED AREA	PHYSICAL AREA
Original spatial overlap	100.00%	Not available
Manual exclusion	0.00%	0.00 ha
Effective analysis footprint	100.00% active	Not available

9. TECHNICAL PROVENANCE & LIMITATIONS

Every metric reported in this assessment is derived directly from verifiable raster artifacts produced during the processing pipeline. No analytical values are interpolated or inferred without evidence.

INDICATOR	OBSERVED VALUE	SPATIAL UNIT	PIPELINE ARTIFACT / METHOD
Wetland source	UNAVAILABLE	class	Wetland mask source selected by wetland context stage
Wetland context status	UNAVAILABLE	status	Wetland mask evaluation result
Effective analysis area	100.00%	active footprint proportion	alignment/effective_analysis_mask.tif
Manual excluded area	0.00%	footprint proportion	exclusions/manual_exclusion_mask.tif
Vegetation degradation	0.00%	% of footprint	degradation/degradation_map.tif (H→M, H→S, M→S)
Vegetation recovery	0.00%	% of footprint	degradation/recovery_map.tif (M→H, S→M, S→H)
CVA Mean Magnitude	0.0000	index units	change/cva_magnitude.tif (||[ΔNDVI, ΔGNDVI, ΔNDRE]||)
Mean ΔNDVI	N/A	index units	change/delta_ndvi.tif (NDVI_T2 – NDVI_T1)
NDWI Water Threshold	0.00	index units	Calibrated Green/NIR NDWI threshold
Physical depletion assessment	NOT_ASSESSED	status	Independent-evidence claim guard

SCIENTIFIC LIMITATIONS & BOUNDARIES

Observation Interval: Remote-sensing comparisons reflect conditions at specific acquisition times. Short-term intervals capture immediate physiological or seasonal responses that may differ from multi-year ecological trends.
Model Inferences: Machine learning vegetation classifications are probabilistic estimates. In-situ field validation is recommended before using findings for regulatory, legal, or compensation purposes.
Spatial Reference: Where imagery lacks a validated Coordinate Reference System (CRS), proportional footprint metrics serve as the authoritative spatial unit. Physical area (hectares) is reported only when georeferencing is confirmed.

Run: run_20260903_section21_example · **Software:** GIS Engine v1.0.0 · **Analysis Basis:** SELECTED_ANALYSIS_FOOTPRINT